Zara Thornton

Contact Information

- Phone: +44 7700 900123
- Email: zara.thornton@email.com
- LinkedIn: linkedin.com/in/zarathornton
- GitHub: github.com/zarathornton

Professional Summary

Highly motivated and results-oriented Machine Learning Research Fellow at Innovate AI Labs with a proven track record in developing and deploying innovative AI/ML solutions. Possessing a strong academic background from Imperial College London and expertise in various programming languages and machine learning frameworks, I am adept at tackling complex problems and delivering impactful results. Passionate about pushing the boundaries of AI research and its applications in real-world scenarios. Seeking a challenging role where I can leverage my skills and experience to contribute to the advancement of AI technology.

Education

- **Imperial College London, London, UK**
  - M.Sc. in Machine Learning, 2021 - 2022
    - GPA: 3.9/4.0
    - Thesis: "Novel Deep Learning Architectures for Time Series Anomaly Detection in Financial Markets"
  - B.Sc. in Computer Science, 2017 - 2021
    - GPA: 3.8/4.0
    - Honours: Dean's List (2018, 2019, 2020, 2021)

Technical Skills

- **Programming Languages:** Python (Pandas, NumPy, SciPy), R, Go
- **Machine Learning Frameworks:** TensorFlow/Keras, PyTorch, Scikit-learn, XGBoost
- **Cloud Computing:** AWS (EC2, S3, Lambda, SageMaker)
- **Databases:** SQL, NoSQL (MongoDB)
- **Data Visualization:** Matplotlib, Seaborn, Tableau
- **Version Control:** Git

Professional Experience

- **Machine Learning Research Fellow, Innovate AI Labs, London, UK** (June 2022 – Present)
    - Developed and implemented novel deep learning models for fraud detection, resulting in a 15% reduction in fraudulent transactions.
    - Conducted extensive research on explainable AI (XAI) techniques and applied them to improve the transparency and interpretability of machine learning models.
    - Collaborated with a team of engineers and researchers to deploy machine learning models to production environments using AWS SageMaker.
    - Presented research findings at international conferences and published a paper in a peer-reviewed journal.
- **Data Science Intern, Quantify Analytics, London, UK** (June 2021 – August 2021)
    - Developed predictive models for customer churn using various machine learning algorithms, improving prediction accuracy by 10%.
    - Performed data cleaning, preprocessing, and feature engineering using Python and R.
    - Created interactive dashboards to visualize data and insights using Tableau.

AI/ML Projects

- **Time Series Anomaly Detection in Financial Markets:** Developed a novel deep learning architecture for detecting anomalies in high-frequency financial data. This project involved extensive data preprocessing, model training, and evaluation using various metrics. The results were presented at the International Conference on Machine Learning (ICML) 2023. (GitHub repository available upon request)
- **Fraud Detection System:** Designed and implemented a real-time fraud detection system using a combination of supervised and unsupervised learning techniques. This project involved working with large datasets, deploying models to a cloud environment, and monitoring model performance in real-time. (Confidential due to client restrictions)
- **Customer Churn Prediction:** Built a predictive model for customer churn using various machine learning algorithms, including logistic regression, support vector machines, and random forests. This project involved data cleaning, feature engineering, model selection, and evaluation. The model

achieved an accuracy of 92% on the test set. (GitHub repository available upon request)